

AI for Communication and Marketing

Mar 25, 2026 - Lab lecture #3: RFM Segmentation





Why Segment Your Customers?

Not all customers are equal. Treating them as if they were leads to wasted budget, irrelevant messaging, and missed opportunities. Segmentation lets you **speak to each group in a language that resonates.**



Precision Targeting

Send the right offer to the right person at the right time



Higher ROI

Focus resources on customers most likely to convert



Better analysis

Having the CB segmented allows to have a deeper understanding of its evolution



DEFINITION

What is RFM Segmentation?

RFM is a **data-driven customer segmentation model** that ranks customers based on three behavioral dimensions derived from transaction history — no surveys, no guesswork.

R — Recency

When did they last buy?

F — Frequency

How often do they buy?

M — Monetary

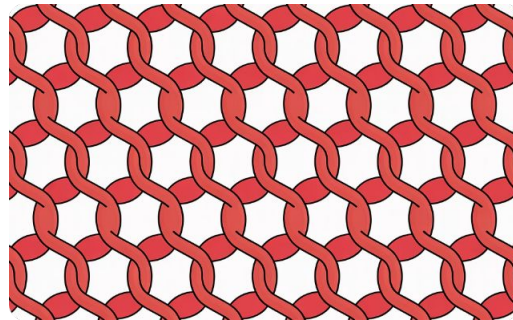
How much do they spend?

The Three Pillars of RFM



Recency

Customers who purchased **recently** are more likely to buy again. A small recency window signals active engagement — they still remember your brand.



Frequency

Repeat buyers are your most **loyal customers**. High frequency indicates satisfaction, habit, and trust — the foundation of brand advocacy.



Monetary Value

Total spend identifies your **highest-value customers**. These are the VIPs who deserve premium attention, exclusive offers, and white-glove treatment.

When to Use RFM

RFM works best when you have **transactional data** — e-commerce, retail, subscriptions, or any repeat-purchase context. It shines in these scenarios:

→ Churn Prevention

Identify customers who haven't purchased in a while and re-engage them before they're gone for good

→ Campaign Prioritization

Allocate marketing budget where it will have the greatest impact based on customer value tiers

→ VIP & Loyalty Programs

Reward your Champions — high R, F, and M scores — with exclusive perks and early access

→ Reactivation Campaigns

Win back "At Risk" or "Hibernating" segments with targeted, time-sensitive incentives

From Data to Actionable Insights



The RFM Workflow

Once scores are assigned (typically 1–5 per dimension), customers fall into named segments — each with a **prescribed marketing action**.

01

Score each dimension

Rank customers 1–5 on R, F, and M independently

02

Combine into a segment label

E.g., 555 = Champion, 111 = Lost

03

Map to a campaign strategy

Each segment gets a tailored communication plan



</> HANDS-ON · PART 1

Calculating RFM Values with Python

We'll use **pandas** to compute the three RFM metrics from a raw transaction dataset.

```
import pandas as pd
from datetime import datetime

# Load transaction data
df = pd.read_csv('transactions.csv', parse_dates=['order_date'])

snapshot_date = datetime(2024, 1, 1)

rfm = df.groupby('customer_id').agg(
    Recency = ('order_date', lambda x: (snapshot_date - x.max()).days),
    Frequency = ('order_id', 'count'),
    Monetary = ('order_value', 'sum')
).reset_index()

print(rfm.head())
```

Assigning RFM Scores and Labels

```
# Score each dimension 1-5
rfm['R_Score'] = pd.qcut(rfm['Recency'],
                        q=5, labels=[5,4,3,2,1])
rfm['F_Score'] = pd.qcut(rfm['Frequency'].rank(method='first'),
                        q=5, labels=[1,2,3,4,5])
rfm['M_Score'] = pd.qcut(rfm['Monetary'],
                        q=5, labels=[1,2,3,4,5])

# Combine into RFM segment string
rfm['RFM_Segment'] = (
    rfm['R_Score'].astype(str) +
    rfm['F_Score'].astype(str) +
    rfm['M_Score'].astype(str)
)
```

```
# Map to human-readable labels
def label_segment(row):
    r, f, m = int(row['R_Score']), int(row['F_Score']),
    int(row['M_Score'])
    if r >= 4 and f >= 4 and m >= 4:
        return 'Champion'
    elif r >= 3 and f >= 3:
        return 'Loyal'
    elif r <= 2:
        return 'At Risk'
    else:
        return 'Potential'

rfm['Label'] = rfm.apply(label_segment, axis=1)
```

What this produces

Each customer now has a **3-digit RFM score** and a human-readable segment label — ready for campaign targeting or deeper analysis.

555
Champion

344
Loyal

211
At Risk

111
Lost

Use **pd.qcut** (quantile-based) to ensure balanced segment sizes across your customer base.

Hands-on Session: Your RFM segmentation challenge

Dataset Information

Additional Information

This is a transactional data set which contains all the transactions occurring between 01/12/2010 and 09/12/2011 for a UK-based and registered non-store online retail. The company mainly sells unique all-occasion gifts. Many customers of the company are wholesalers.

Variables Table

Variable Name	Role	Type	Description
InvoiceNo	ID	Categorical	a 6-digit integral number uniquely assigned to each transaction. If this code starts with letter 'c', it indicates a cancellation
StockCode	ID	Categorical	a 5-digit integral number uniquely assigned to each distinct product
Description	Feature	Categorical	product name
Quantity	Feature	Integer	the quantities of each product (item) per transaction
InvoiceDate	Feature	Date	the day and time when each transaction was generated
UnitPrice	Feature	Continuous	product price per unit
CustomerID	Feature	Categorical	a 5-digit integral number uniquely assigned to each customer
Country	Feature	Categorical	the name of the country where each customer resides

Hands-on Session: Your RFM segmentation challenge

01

Dataset Audit

Begin with a comprehensive audit of `Online_Retail.xlsx` to understand the structure and key characteristics.

03

Build an RFM Segmentation

Create a clean dataset with the features you need to calculate RFM variables. Then build the segments and name them based on their characteristics.

02

Data Quality: Cleaning Strategy

Identify and address missing values and outliers. Justify your cleaning choices based on feature distributions and potential impact on models.

04

Insights and actions for the CEO

Present your key findings and actionable recommendations with appropriate campaign strategy for the CEO, with clear slides.